



Business Requirements Document (BRD)

1. Document Control

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Document Purpose

This document defines the business requirements for the digital transformation of Green Valley Medical Center. It consolidates the key business needs, requirements, scope, and expected outcomes identified throughout the Business Analysis process.

The BRD covers the requirements associated with patient management, appointment management, electronic medical records, laboratory workflows, payment processes, follow-up management, reporting, and information integration within the clinic.

2. Executive Summary

2.1 Project Overview

Green Valley Medical Center is a fictional multi-specialty healthcare clinic used as the business scenario for this portfolio project. The clinic is undergoing a simulated digital transformation initiative to improve the management and integration of clinical and administrative processes.

2.2 Business Need

The current environment relies on fragmented systems and manual processes across patient management, appointments, payments, medical records, laboratory workflows, and operational reporting. This creates duplicated data entry, limited information visibility, communication gaps, and delays in accessing relevant patient and operational information.

2.3 Proposed Direction

The proposed future state introduces a more integrated digital environment in which patient, appointment, clinical, laboratory, payment, and reporting information can be managed through connected processes and centralized records.

2.4 Expected Outcomes

The initiative is expected to:

- Reduce manual and duplicated processes
- Improve access to patient and operational information

- Improve communication between departments
 - Support more consistent clinical and administrative workflows
 - Improve reporting and data visibility
 - Support a better patient experience
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3. Business Background

Aqui vamos dar **contexto suficiente para alguém que não acompanhou as outras documentações** entender o ambiente de negócio. Não precisamos recontar o Current State inteiro.

3.1 Organisation Overview

Green Valley Medical Center is a fictional multi-specialty healthcare clinic providing outpatient medical services across multiple departments. The clinic involves several operational and clinical teams that collaborate throughout the patient journey, including Reception, Healthcare Providers, and Laboratory staff.

3.2 Current Business Environment

The clinic currently operates across multiple processes and systems to manage patient registration, appointments, referrals, payments, consultations, medical records, laboratory requests, and reporting.

Information is distributed across different systems and manual records, requiring staff to perform activities such as data entry, information retrieval, communication between departments, and reporting through fragmented processes.

3.3 Key Business Areas

The project focuses on the following areas:

- Patient registration and management
- Appointment scheduling and follow-up
- Medical referrals
- Payment management

- Electronic medical records
- Laboratory requests and results
- Operational and financial reporting
- Healthcare provider payments

3.4 Business Context

The initiative considers the needs of both clinical and administrative users. The proposed transformation aims to create a more integrated working environment while maintaining appropriate access to patient information according to user responsibilities.

4. Business Problem

4.1 Problem Statement

Green Valley Medical Center currently relies on fragmented systems and manual processes to manage key clinical and administrative activities.

This creates duplicated data entry, inconsistent information, delays in accessing patient records, communication gaps between departments, and limited visibility into operational and financial performance.

4.2 Business Impact

The current environment may result in:

- Increased administrative workload and manual rework
- Delays in patient and clinical workflows
- Higher risk of data inconsistencies and errors
- Limited visibility across departments
- Manual effort required for reporting and payment tracking
- Communication gaps between clinical and administrative teams
- Potential negative impact on the patient experience

4.3 Business Need

The clinic requires a more integrated approach to managing patient, appointment, clinical, laboratory, payment, and reporting information.

The proposed transformation aims to reduce fragmentation, improve information visibility, support more consistent processes, and enable staff to access relevant information according to their roles and responsibilities.

5. Business Objectives

5.1 Primary Objective

The primary objective of the digital transformation initiative is to improve the integration, efficiency, and visibility of clinical and administrative processes across Green Valley Medical Center.

5.2 Specific Objectives

The initiative aims to:

- Improve the management of patient information and medical records
- Reduce manual and duplicated data entry
- Improve appointment scheduling and follow-up management
- Integrate clinical and laboratory information
- Improve payment tracking and financial reporting
- Provide better visibility into operational and clinical performance
- Support secure, role-appropriate access to patient information
- Improve communication and coordination between departments
- Support more consistent processes across the patient journey
- Improve the overall patient experience

5.3 Expected Business Value

The expected business value includes increased operational efficiency, improved data quality, better access to information, reduced administrative effort, and stronger support for clinical and management decision-making.

6. Project Scope

6.1 In Scope

The project includes the analysis, design, development, testing, and simulated implementation of the digital transformation of the clinic's core clinical and administrative processes.

The following areas are included within the project scope:

- Patient registration and patient information management
- Appointment scheduling and follow-up management
- Medical referral management
- Payment status and payment tracking
- Electronic medical records
- Laboratory requests and results
- Clinical and administrative information integration
- Operational, clinical, and financial reporting
- Healthcare provider payment reporting
- Generation of medical documents
- Role-based access to patient and operational information
- Data validation and consistency
- Auditability of relevant system activities

6.2 Out of Scope

The following activities are outside the scope of this project:

- Procurement of software or hardware
- Replacement or installation of physical medical equipment
- Changes to clinical treatment protocols
- Direct clinical decision-making
- Patient-facing mobile application development
- Migration of real patient data into a live production environment

6.3 Scope Boundary

The project covers the simulated end-to-end transformation of Green Valley Medical Center, from business analysis and requirements definition through solution design, development and configuration, testing, and simulated deployment.

The Business Analysis phase focuses on understanding the current environment, defining the future state, identifying business and system requirements, modelling processes, and defining the expected solution capabilities.

Later project phases will address solution design, database design, development and configuration, testing, deployment, and project closure.

Because Green Valley Medical Center is a fictional organisation, the implementation and deployment activities are simulated for portfolio purposes and do not involve a real production

environment or real patient data.

7. Stakeholders

7.1 Key Stakeholders

The project involves clinical, administrative, management, and technical stakeholders who interact with or are affected by the proposed digital transformation.

Key stakeholders include:

- Clinic Director – Project Sponsor
- Project Manager
- Business Analyst
- Reception Team
- Healthcare Providers
- Laboratory Staff
- Finance Team
- Administrative Staff
- IT Support / Solution Provider
- Patients

7.2 Stakeholder Involvement

Stakeholders contribute to the project according to their responsibilities and level of involvement.

Clinical and administrative users provide business and workflow knowledge, while management provides strategic direction and project oversight. The Business Analyst is responsible for eliciting, analysing, and documenting requirements and supporting communication between stakeholders and the implementation team.

8. Current State Summary

8.1 Current Environment

The current environment relies on fragmented systems and manual processes to manage patient records, appointments, billing, laboratory results, and other operational activities.

Information is distributed across different systems and records, with communication between teams relying on processes such as phone calls and email.

8.2 Key Current State Challenges

The main challenges identified are:

- Fragmented information across different systems
 - Manual and duplicated data entry
 - Limited visibility across departments
 - Communication gaps between teams
 - Difficulty accessing complete patient information
 - Manual reporting and administrative activities
 - Limited integration between clinical and administrative processes
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9. Future State Summary

9.1 Future State Vision

The proposed future state introduces a more integrated digital environment for managing the clinic's clinical and administrative processes.

Patient, appointment, clinical, laboratory, payment, and reporting information will be managed through connected processes and centralized records.

9.2 Expected Future State Capabilities

The future state is expected to support:

- Centralized patient information
- Integrated appointment management
- Electronic medical records
- Integrated laboratory information
- Payment and financial tracking
- Follow-up appointment management
- Operational and financial reporting
- Role-based access to information
- Improved information visibility across departments

These capabilities are reflected in the defined functional and non-functional requirements.

10. Business Requirements

10.1 Patient Management

BR-01 — Integrated Patient Management

The clinic requires a centralized approach to managing patient information, including registration data, referrals, previous appointments, healthcare providers, and relevant clinical history.

10.2 Appointment Management

BR-02 — Integrated Appointment Management

The clinic requires an integrated process for scheduling, managing, classifying, and following up on patient appointments.

10.3 Clinical Information Management

BR-03 — Electronic Clinical Records

The clinic requires electronic management of patient clinical information, including previous clinical records, treatment information, and consultation documentation.

10.4 Laboratory Management

BR-04 — Integrated Laboratory Workflow

The clinic requires an integrated process for requesting, tracking, storing, and accessing laboratory test results.

10.5 Payment Management

BR-05 — Payment Visibility

The clinic requires visibility of consultation payment status and financial information associated with services provided.

10.6 Reporting

BR-06 — Operational and Financial Reporting

The clinic requires reporting capabilities to support operational, demographic, revenue, laboratory, and healthcare provider payment analysis.

10.7 Information Integration

BR-07 — Centralized Information

The clinic requires relevant clinical and operational information to be managed through an integrated information environment.

10.8 Security and Privacy

BR-08 — Secure Information Access

The clinic requires patient information and system functionality to be protected through appropriate access controls, authentication, data protection, and auditability.

11. Functional Requirements

11.1 Requirements Overview

The functional requirements define the capabilities that the proposed system must provide to support the identified business needs.

The requirements cover:

- Patient registration and patient history
- Appointment scheduling and classification
- Follow-up scheduling
- Referral verification
- Payment status
- Electronic medical records
- Previous clinical records
- Laboratory requests and results
- Operational and financial reporting
- Healthcare provider payment reporting
- Medical document generation
- Centralized information

The detailed functional requirements are documented separately as **FR-01 to FR-19**.

12. Non-Functional Requirements

12.1 Requirements Overview

The non-functional requirements define the quality, security, performance, usability, and operational characteristics expected from the proposed solution.

The requirements cover:

- Role-based access control
- Authentication

- Data protection
- Patient data privacy
- Data validation
- Data consistency
- System response time
- System availability
- User-friendly interface
- Audit trail

The detailed non-functional requirements are documented separately as **NFR-01 to NFR-10**.

13. Business Rules

13.1 Patient and Referral Rules

- Patient information must be registered by authorized staff.
- Required medical referrals must be recorded and verified during the patient check-in process before the check-in is completed.
- Patient clinical information must only be accessible to authorized users.

13.2 Appointment Rules

- Appointments must be managed through the integrated scheduling process.
- Appointments must be distinguishable by department, appointment type, and follow-up status.
- A follow-up appointment may be scheduled by Reception when requested by the healthcare provider.

13.3 Clinical Information Rules

- Healthcare providers may access and update electronic medical records according to their authorized role.
- Previous clinical records must remain available to authorized healthcare providers.
- Laboratory results must be associated with the relevant patient record.

13.4 Payment Rules

- The system must provide visibility of whether a scheduled consultation has been paid.
- Financial information must only be accessible to users with appropriate responsibilities.

These rules are derived from the defined functional and non-functional requirements and are intended to guide the future solution design.

14. Assumptions & Constraints

14.1 Assumptions

The project assumes that:

- Relevant stakeholders will be available to provide business and workflow information.
- Business users will participate in requirements validation and review.
- The proposed solution will support the identified clinical and administrative workflows.
- Appropriate access permissions will be defined according to user responsibilities.
- Existing patient and operational information can be structured for use within the proposed solution.

14.2 Constraints

The project is constrained by:

- The need to protect sensitive patient information.
 - The complexity of integrating clinical and administrative processes.
 - The requirement for appropriate role-based access.
 - Potential limitations associated with existing data quality and fragmented information.
 - The project scope being limited to the defined business and system requirements.
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15. Risks & Dependencies

15.1 Key Risks

The project should consider the following risks:

- Poor data quality may affect the reliability of migrated or consolidated information.
- Resistance to change may affect adoption of new processes.
- Inadequate access controls may create privacy and security risks.
- Integration challenges may affect information flow between clinical and administrative processes.
- Insufficient stakeholder involvement may result in incomplete or misunderstood requirements.

15.2 Key Dependencies

Successful delivery depends on:

- Stakeholder participation and requirements validation
 - Availability and quality of existing business data
 - Appropriate system configuration and integration capabilities
 - Definition of user roles and access permissions
 - User acceptance of the proposed future-state processes
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16. Success Criteria

16.1 Business Success Criteria

The project will be considered successful when the proposed solution is capable of:

- Supporting the defined patient management processes
- Supporting integrated appointment and follow-up management
- Providing authorized healthcare providers with appropriate access to clinical information
- Supporting laboratory request and result management
- Providing visibility into payment and financial information
- Supporting the required operational and management reporting
- Providing appropriate protection of patient information
- Reducing reliance on fragmented and manual processes

16.2 User and Operational Success

Success will also be evaluated based on whether clinical and administrative users can perform their relevant activities through clear, consistent, and integrated processes with appropriate access to the information required for their roles. Detailed project success criteria and performance targets are defined in the Project Success Criteria document.

17. Traceability

17.1 Requirements Traceability

Requirements traceability establishes the relationship between business needs, business requirements, and system requirements.

Business Requirement	Functional Requirements	Non-Functional Requirements
BR-01 — Integrated Patient Management	FR-01, FR-02, FR-06	NFR-05, NFR-06

BR-02 — Integrated Appointment Management	FR-03, FR-04, FR-05, FR-06	NFR-05, NFR-06, NFR-09
BR-03 — Electronic Clinical Records	FR-08, FR-09, FR-18	NFR-01, NFR-03, NFR-04, NFR-10
BR-04 — Integrated Laboratory Workflow	FR-10, FR-11, FR-12	NFR-06
BR-05 — Payment Visibility	FR-07	NFR-01, NFR-06
BR-06 — Operational and Financial Reporting	FR-13, FR-14, FR-15, FR-16, FR-17	NFR-06, NFR-09
BR-07 — Centralized Information	FR-19	NFR-06, NFR-07, NFR-08
BR-08 — Secure Information Access	FR-01, FR-08, FR-09, FR-19	NFR-01, NFR-02, NFR-03, NFR-04, NFR-10

17.2 Traceability Purpose

The traceability structure provides a link between the identified business needs and the detailed system requirements.

It can be used throughout the project to support requirements validation, impact analysis, testing, and change management.
